

Worksheet - 2

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Subject Name	: Business Analytics	Subject Code	: 25CAH-703

1. Aim/Overview of the practical:

- Create a Pivot table to quickly summarize large databases, Group within Pivot Tables.
- Create multiple Pivot Tables on a single worksheet.

2. Objective:

- To understand the concept and use of Pivot Tables in MS Excel.
- To summarize and organize product sales data efficiently.
- To analyze sales based on region, product, payment method, customer type, and promotion.
- To learn how to use Count, Sum, and Average functions in Pivot Tables for calculating total sales, number of orders, average sales, and total profit.
- To calculate and analyze Mean and Median of sales data using Excel functions.
- To compare total sales, quantity sold, unit price, and profit across different regions and products.
- To create Pivot Charts for visual analysis of product sales data.

3. Apparatus Used:

- Microsoft Excel

4. Procedure:

- Open the Excel file containing the sales data.
- Go to Sheet1, which contains the original dataset.
- Select any cell inside the dataset.
- Press Ctrl + A to select the complete data range.
- Go to Insert → PivotTable.
- Select From Table/Range if required.
- Choose New Worksheet and click OK.
- The PivotTable Fields pane will appear on the right side.
- To summarize sales according to Region:
 - Drag Region → Rows.
 - Drag TotalPrice → Values.
- Excel will automatically calculate Sum of TotalPrice for each region.
- To calculate the number of sales/orders:
 - Drag OrderID → Values.
 - Change it to Count of OrderID if Excel selects Sum.
- The Pivot Table will now show the total sales and number of orders for each region.

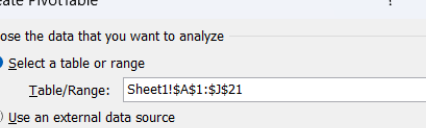
5. Pivot Table Creation:

Step 1: Enter the Data: Open Excel and enter the sales dataset with these columns:

J1		TotalPrice																						
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
	Date	Region	Product	Quantity	UnitPrice	StoreLocation	CustomerType	Discount	Salesperson	TotalPrice	PaymentMethod	Promotion	Returned	OrderID	CustomerName	ShippingCost	OrderDate	DeliveryDate	SalesManager					
1	2023-02-23	East	Laptop	14	163.6	Store B	Wholesale	0	Eva	2290.4	Online	FREESHIP	0	REG10000Cust 6183	43.34		2023-02-23	2023-02-27	Eric					
2	2024-12-19	South	Phone	1	51.01	Store A	Retail	0	Alice	54.01	Gift Card	SAVE10	1	REG10000Cust 2584	5.3		2024-12-19	2024-12-28	Sophie					
3	2023-05-10	North	Desk	14	346.18	Store B	Wholesale	0	Eva	4361.968	Online	WINTER15	0	REG10000Cust 5998	20.46		2023-05-10	2023-05-10	Wendy					
4	2025-02-26	Central	Chair	18	384.82	Store A	Wholesale	0.15	Frank	5887.746	Gift Card	FREESHIP	0	REG10000Cust 7136	27.95		2025-02-26	2025-03-02	Cameron					
5	2023-06-24	East	Desk	18	237.76	Store C	Retail	0	Carlos	4279.68	Online	SAVE10	0	REG10000Cust 6506	5.73		2023-06-24	2023-06-27	Eric					
7	2024-02-20	Central	Monitor	2	385.09	Store D	Wholesale	0.15	Diana	654.653	Credit Card		0	REG10000Cust 3909	11.92		2024-02-20	2024-03-01	Cameron					
8	2023-01-11	South	Chair	7	17.5	Store C	Retail	0.1	Alice	110.25	Debit Card	WINTER15	0	REG10000Cust 7887	5.02		2023-01-11	2024-01-14	Sophie					
9	2023-01-09	Central	Chair	3	330.22	Store C	Wholesale	0.15	Carlos	842.061	Cash		0	REG10000Cust 5301	48.01		2023-01-09	2023-01-11	Cameron					
10	2024-10-16	South	Chair	3	432.04	Store B	Wholesale	0.05	Bob	1231.314	Debit Card	FREESHIP	1	REG10000Cust 2284	42.11		2024-10-16	2024-10-18	Sophie					
11	2025-03-05	West	Desk	5	323.28	Store D	Retail	0	Frank	1654.76	Gift Card	SAVE10	0	REG10000Cust 3732	41.1		2025-03-05	2025-03-13	Wendy					
12	2024-04-10	East	Phone	15	523.29	Store B	Wholesale	0.15	Eva	6671.94	Debit Card		0	REG10000Cust 1462	12.14		2024-04-10	2024-04-10	Ryan					
13	2024-07-12	East	Phone	19	207.35	Store D	Retail	0.1	Eva	3545.685	Gift Card		1	REG10001Cust 4330	44.89		2024-07-12	2024-07-18	Eric					
14	2023-07-21	Central	Phone	1	120.53	Store B	Retail	0.05	Diana	114.5035	Cash	WINTER15	0	REG10001Cust 2372	20.35		2023-07-21	2023-07-31	Cameron					
15	2025-01-18	West	Tablet	1	157.83	Store A	Wholesale	0.15	Carlos	134.1555	Debit Card		0	REG10001Cust 4672	42.82		2025-01-18	2025-01-22	Wendy					
16	2024-12-08	North	Desk	15	220.3	Store B	Retail	0.15	Diana	2808.825	Debit Card		0	REG10001Cust 4548	39.66		2024-12-08	2024-12-15	Ryan					
17	2025-05-11	West	Chair	17	162.96	Store A	Wholesale	0.05	Eva	2631.804	Gift Card	FREESHIP	0	REG10001Cust 7037	8.11		2025-05-11	2025-05-17	Wendy					
18	2023-02-26	West	Laptop	19	236.11	Store D	Wholesale	0	Bob	4486.09	Online	WINTER15	0	REG10001Cust 4975	34.11		2023-02-26	2023-03-04	Wendy					
19	2024-05-28	Central	Desk	11	573.14	Store D	Retail	0.05	Diana	5989.313	Cash	SAVE10	0	REG10001Cust 8014	34.71		2024-05-28	2024-06-01	Cameron					
20	2023-11-30	East	Phone	20	519.04	Store B	Retail	0.15	Carlos	8823.68	Gift Card	FREESHIP	0	REG10001Cust 1228	19.43		2023-11-30	2023-12-08	Eric					
21	2023-06-22	West	Desk	14	30.83	Store B	Retail	0	Carlos	431.62	Online	WINTER15	1	REG10001Cust 4247	7.1		2023-06-22	2023-07-02	Wendy					
22	2025-06-05	West	Phone	10	319.05	Store D	Wholesale	0	Diana	3190.5	Online		0	REG10002Cust 4199	28.43		2025-06-05	2025-06-08	Wendy					
23	2023-03-03	Central	Chair	14	312.79	Store A	Retail	0.05	Bob	4160.107	Online		1	REG10002Cust 4186	46.83		2023-03-03	2023-03-08	Cameron					
24	2024-02-18	North	Laptop	14	395.45	Store C	Retail	0	Carlos	5536.3	Cash	FREESHIP	1	REG10002Cust 8137	36.45		2024-02-18	2024-02-22	Ryan					
25	2023-08-06	Central	Laptop	12	472.72	Store A	Retail	0.15	Diana	4821.744	Credit Card		0	REG10002Cust 4638	27.97		2023-08-06	2023-08-14	Cameron					
26	2024-05-23	North	Desk	5	288.01	Store D	Wholesale	0	Eva	1440.05	Credit Card		0	REG10002Cust 8538	46.69		2024-05-23	2024-05-30	Ryan					
27	2023-10-23	East	Phone	13	562.73	Store D	Retail	0	Diana	7509.77	Gift Card	SAVE10	0	REG10002Cust 6322	38.13		2023-10-23	2023-10-28	Eric					
28	2024-06-16	West	Chair	1	106.79	Store B	Retail	0.05	Diana	106.79	Debit Card	WINTER15	0	REG10002Cust 2609	11.59		2024-06-16	2024-06-16	Wendy					
29	2023-03-03	North	Phone	4	279.49	Store B	Retail	0	Diana	1117.96	Gift Card	FREESHIP	0	REG10002Cust 6880	24.78		2023-03-03	2023-03-08	Ryan					
30	2024-11-18	South	Laptop	1	178.55	Store A	Retail	0.15	Carlos	151.7675	Debit Card	SAVE10	1	REG10002Cust 1997	7.61		2024-11-18	2024-11-21	Sophie					
31	2023-05-09	Central	Tablet	1	472.07	Store A	Wholesale	0.1	Carlos	424.863	Online	FREESHIP	0	REG10002Cust 7072	33.17		2023-05-09	2023-05-17	Cameron					
32	2024-12-07	North	Tablet	12	306.29	Store D	Retail	0	Frank	3675.48	Debit Card	SAVE10	0	REG10003Cust 7201	37.38		2024-12-07	2024-12-14	Ryan					
33	2024-02-13	North	Phone	7	434.77	Store C	Retail	0.15	Frank	2586.8815	Online	SAVE10	0	REG10003Cust 9923	12.35		2024-02-13	2024-02-15	Ryan					
34	2025-03-09	South	Desk	6	64.98	Store C	Retail	0	Carlos	389.88	Debit Card	WINTER15	0	REG10003Cust 2205	21.25		2025-03-09	2025-03-15	Sophie					
35	2023-04-05	Central	Desk	17	476.49	Store D	Retail	0	Alice	8100.33	Gift Card		0	REG10003Cust 6403	17.91		2023-04-05	2023-04-11	Cameron					
36	2025-05-08	North	Monitor	4	559.72	Store C	Retail	0.15	Diana	1960.048	Online	SAVE10	0	REG10003Cust 3953	41.81		2025-05-08	2025-05-08	Ryan					
37	2023-09-03	North	Tablet	14	82.54	Store C	Retail	0	Alice	1149.56	Credit Card		0	REG10003Cust 7902	12.57		2023-09-03	2023-09-11	Ryan					
38	2025-03-21	Central	Chair	9	343.21	Store A	Retail	0.05	Carlos	2334.4455	Debit Card	FREESHIP	0	REG10003Cust 9968	40.29		2025-03-21	2025-03-28	Cameron					

Step 2: Select the Data

1. Select the complete dataset including the headings.
2. Go to the Insert tab.
3. Click PivotTable.
4. Choose the location where the PivotTable should be placed.



The screenshot shows the 'Create PivotTable' dialog box. The 'Table/Range' is set to 'Sheet1!\$A\$1:\$J\$21'. The 'Existing Worksheet' option is selected, and the 'Location' is set to 'Sheet1!\$B\$26'. The 'OK' and 'Cancel' buttons are at the bottom.

Create PivotTable

Choose the data that you want to analyze

☒ Select a table or range

Table/Range: Sheet1!\$A\$1:\$J\$21

☐ Use an external data source

Choose Connection...

Connection name:

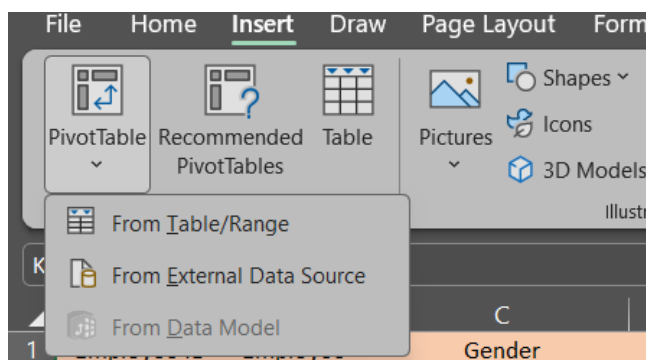
Choose where you want the PivotTable report to be placed

☐ New Worksheet

☒ Existing Worksheet

Location: Sheet1!\$B\$26

OK Cancel



1. Insert – Pivot Table
2. Select the location to place the pivot table

Pivot Table 1 – Region-wise Total Sales

Select the complete Product Sales dataset.

Insert $\rightarrow$ PivotTable.

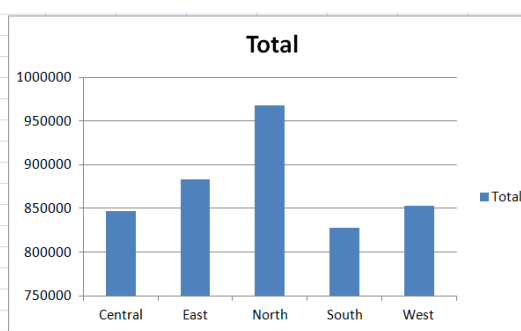
Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	—	Sum of TotalPrice	—

3	Row Labels	Sum of TotalPrice
4	Central	847153.6845
5	East	883633.718
6	North	967957.98
7	South	827768.187
8	West	853478.859
9	Grand Total	4379992.429
10		
11		
12		
13		
14		
15		
16		
17		
18		



PivotTable Field List

Choose fields to add to report:

- ☐ Date
- ☒ Region
- ☐ Product
- ☐ Quantity
- ☐ UnitPrice
- ☐ StoreLocation
- ☐ CustomerType
- ☐ Discount
- ☐ Salesperson
- ☒ TotalPrice
- ☐ PaymentMethod
- ☐ Promotion
- ☐ Returned
- ☐ OrderID
- ☐ CustomerName
- ☐ ShippingCost

Drag field between areas below:

☒ Report Filter

☐ Column Labels

☐ Row Labels

☐ Values

Pivot Table 2 – Product-wise Total Sales

Select the complete Product Sales dataset.

Insert → PivotTable.

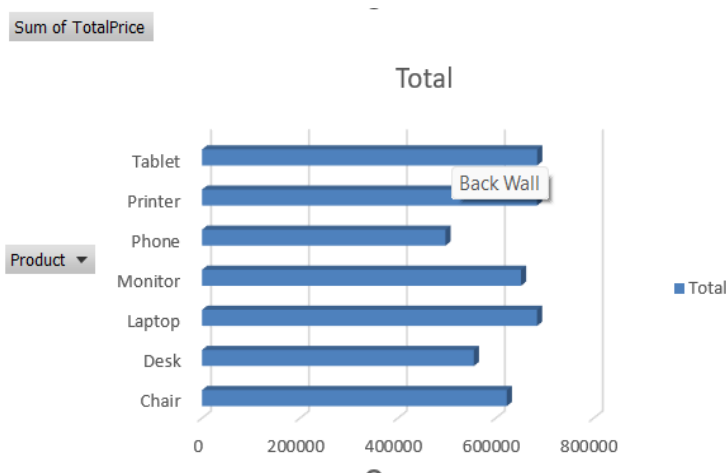
Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Product	—	Sum of TotalPrice	—

Row Labels	Sum of TotalPrice
Chair	622589.478
Desk	555266.6645
Laptop	684417.2415
Monitor	651629.3925
Phone	497162.851
Printer	684387.4155
Tablet	684539.3855
Grand Total	4379992.429



Pivot Table 3 – Product-wise Total Quantity

Select the complete Product Sales dataset.

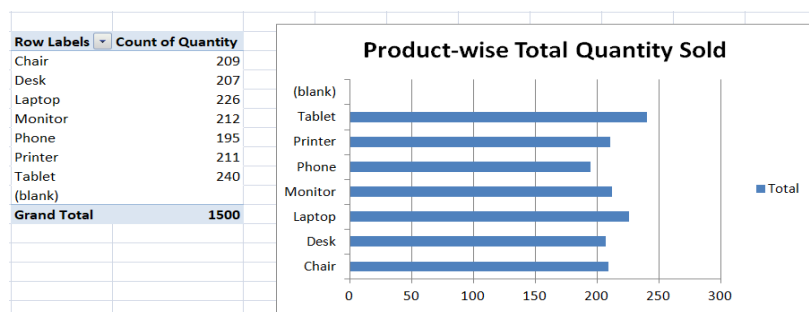
Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Product	—	Sum of Quantity	—



Drag fields between areas below:

Report Filter	Legend Fields (Series)
Axis Fields (Categories)	Σ Values
Product	Count of Quantity

Pivot Table 4 – StoreLocation-wise Total Sales

Select the complete Product Sales dataset.

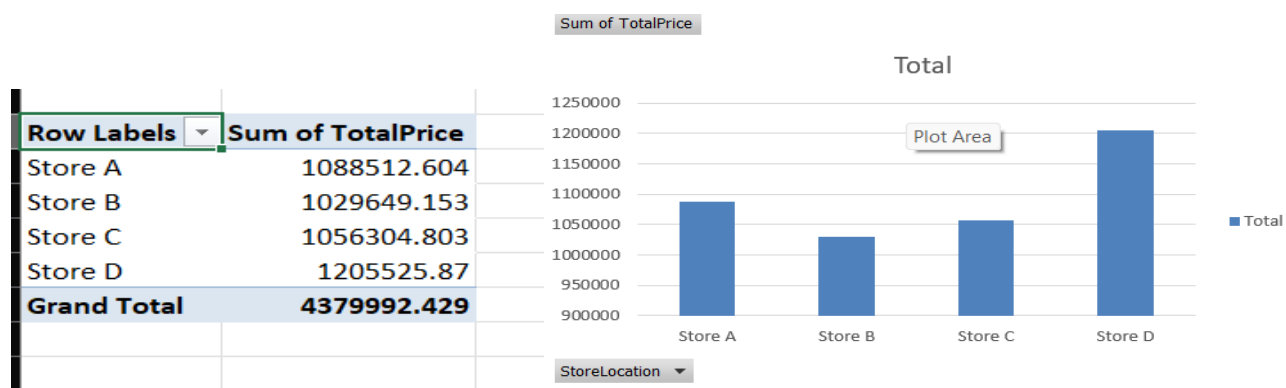
Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
StoreLocation	—	Sum of TotalPrice	—



Pivot Table 5 – CustomerType-wise Total Sales

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
CustomerType	—	Sum of TotalPrice	—

Row Labels	Sum of TotalPrice
Retail	2195525.593
Wholesale	2184466.836
Grand Total	4379992.429

Pivot Table 6 – Salesperson-wise Total Sales

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

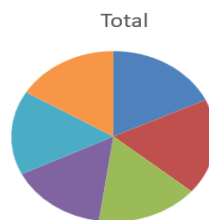
Rows	Columns	Values	Filter
Salesperson	—	Sum of TotalPrice	—



Sum of TotalPrice

Row Labels	Sum of TotalPrice
Alice	786165.84
Bob	796780.5215
Carlos	707166.716
Diana	676568.2495
Eva	698669.5795
Frank	714641.522
Grand Total	4379

Sum of TotalPrice
Value: 714641.522
Row: Frank



Salesperson
 ■ Alice
 ■ Bob
 ■ Carlos
 ■ Diana
 ■ Eva
 ■ Frank

Chart Area

Pivot Table 7 – PaymentMethod-wise Total Sales

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
PaymentMethod	—	Sum of TotalPrice	—

Row Labels	Sum of TotalPrice
Cash	950387.849
Credit Card	866483.3745
Debit Card	770420.664
Gift Card	821585.391
Online	971115.15
Grand Total	4379992.429

Pivot Table 8 – Promotion-wise Total Sales

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Promotion	—	Sum of TotalPrice	—

Row Labels	Sum of TotalPrice
FREESHIP	1238076.924
SAVE10	989656.8055
WINTER15	1078011.928
(blank)	1074246.772
Grand Total	4379992.429

Pivot Table 9 – Quantity by Region and Product

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	Product	Sum of Quantity	—

Sum of Quantity	Column Labels								
Row Labels	Chair	Desk	Laptop	Monitor	Phone	Printer	Tablet	Grand Total	
Central	381	433	485	446	290	489	578	3102	
East	378	398	528	364	539	456	415	3078	
North	489	581	418	523	353	515	563	3442	
South	482	256	544	345	495	446	385	2953	
West	392	358	433	499	293	430	636	3041	
Grand Total	2122	2026	2408	2177	1970	2336	2577	15616	

Pivot Table 10 – Sales by Region and CustomerType

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	CustomerType	Sum of TotalPrice	—

Sum of TotalPrice	Column Labels		
Row Labels	Retail	Wholesale	Grand Total
Central	474462.4275	372691.257	847153.6845
East	417764.449	465869.269	883633.718
North	522368.1545	445589.8255	967957.98
South	381137.4345	446630.7525	827768.187
West	399793.1275	453685.7315	853478.859
Grand Total	2195525.593	2184466.836	4379992.429



Pivot Table 11 – Sales by Product and PaymentMethod

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Product	PaymentMethod	Sum of TotalPrice	—

Sum of TotalPrice	Column Labels					
Row Labels	Cash	Credit Card	Debit Card	Gift Card	Online	Grand Total
Chair	124745.6605	144091.4265	92609.533	123215.833	137927.025	622589.478
Desk	122131.4895	131365.069	96592.109	108064.0665	97113.9305	555266.6645
Laptop	160219.1005	151552.7195	113786.9485	94855.969	164002.504	684417.2415
Monitor	134153.6725	130644.4435	106056.524	129015.519	151759.2335	651629.3925
Phone	94110.919	88010.202	102989.217	89684.241	122368.272	497162.851
Printer	182035.614	89367.063	122631.486	121895.7315	168457.521	684387.4155
Tablet	132991.393	131452.451	135754.8465	154854.031	129486.664	684539.3855
Grand Total	950387.849	866483.3745	770420.664	821585.391	971115.15	4379992.429

Pivot Table 12 – Average UnitPrice by Product

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Product	—	Average of UnitPrice	—

Row Labels	Average of UnitPrice
Chair	319.7035407
Desk	295.180628
Laptop	303.7332743
Monitor	317.2966509
Phone	271.1922564
Printer	301.9333175
Tablet	282.579
Grand Total	298.8269467

Pivot Table 13 – Average Discount by Region

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	—	Average of Discount	—

Row Labels	Sum of Discount
Central	22.35
East	22.2
North	23.35
South	21.6
West	20.2
Grand Total	109.7

Pivot Table 14 – Shipping Cost by Region

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	—	Sum of ShippingCost	—

Row Labels	Sum of ShippingCost
Central	8394.56
East	8795.31
North	8003
South	8391.53
West	7676.54
Grand Total	41260.94

Pivot Table 15 – Order Count by Region

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	—	Count of OrderID	—

Row Labels	Count of OrderID
Central	301
East	311
North	309
South	295
West	284
Grand Total	1500

Pivot Table 16 – Order Count by CustomerType

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
CustomerType	—	Count of OrderID	—

Row Labels	Count of OrderID
Retail	744
Wholesale	756
Grand Total	1500

Pivot Table 17 – Returned Orders by Region

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	Returned	Count of OrderID	—

Count of OrderID	Column Labels		
Row Labels	0	1	Grand Total
Central	231	70	301
East	230	81	311
North	239	70	309
South	218	77	295
West	210	74	284
Grand Total	1128	372	1500

Pivot Table 18 – Sales by Salesperson and Region

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Salesperson	Region	Sum of TotalPrice	—

Sum of TotalPrice	Column Labels					
Row Labels	Central	East	North	South	West	Grand Total
Alice	155527.42	163494.8505	114607.5655	153319.7935	199216.2105	786165.84
Bob	146204.123	194771.844	168017.591	154037.6065	133749.357	796780.5215
Carlos	156260.9495	126261.6435	175749.3075	116629.032	132265.7835	707166.716
Diana	122602.544	131527.0415	144354.9705	148285.5005	129798.193	676568.2495
Eva	154827.6415	156856.0745	147226.568	107164.8505	132594.445	698669.5795
Frank	111731.0065	110722.264	218001.9775	148331.404	125854.87	714641.522
Grand Total	847153.6845	883633.718	967957.98	827768.187	853478.859	4379992.429

Pivot Table 19 – Sales by Region and Promotion

Select the complete Product Sales dataset.

Insert → PivotTable.

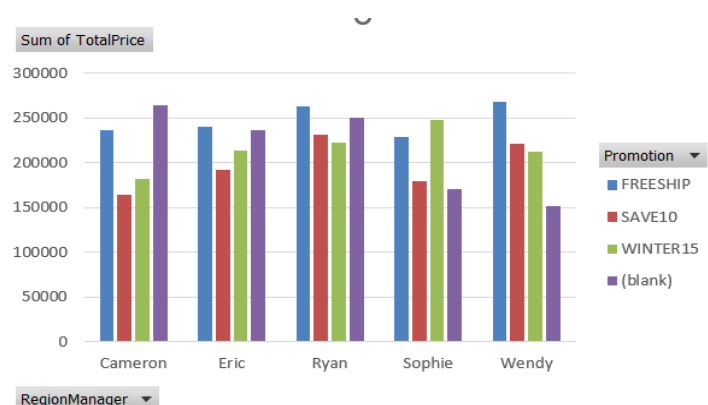
Choose New Worksheet.

Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
Region	Promotion	Sum of TotalPrice	—

Sum of TotalPrice	Column Labels				
Row Labels	FREESHIP	SAVE10	WINTER15	(blank)	Grand Total
Central	236781.8175	164366.9305	181470.881	264534.0555	847153.6845
East	240842.5965	192640.798	213267.73	236882.5935	883633.718
North	262989.915	231533.4765	223181.681	250252.9075	967957.98
South	229037.135	180025.5635	248086.5295	170618.959	827768.187
West	268425.4595	221090.037	212005.1065	151958.256	853478.859
Grand Total	1238076.924	989656.8055	1078011.928	1074246.772	4379992.429



Pivot Table 20 – Sales and Quantity by RegionManager

Select the complete Product Sales dataset.

Insert → PivotTable.

Choose New Worksheet.

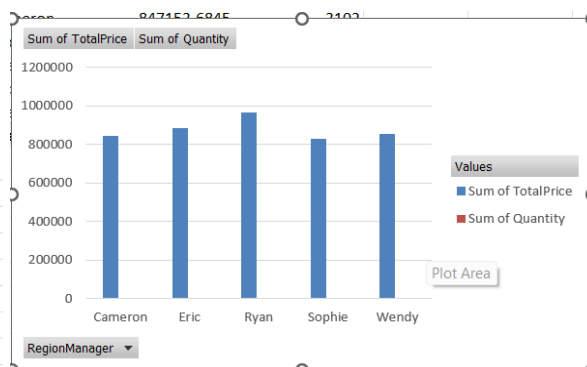
Drag the fields as below.

Read the PivotTable to answer the question.

Rows	Columns	Values	Filter
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RegionManager	—	Sum of TotalPrice; Sum of Quantity	—
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Row Labels	Sum of TotalPrice	Sum of Quantity
Cameron	847153.6845	3102
Eric	883633.718	3078
Ryan	967957.98	3442
Sophie	827768.187	2953
Wendy	853478.859	3041
Grand Total	4379992.429	15616



Learning Outcomes:

1. I learned how to create and use Pivot Tables and Pivot Charts in MS Excel.
2. I learned how to organize and summarize product sales data based on region, product, payment method, customer type, and promotion.
3. I learned how to use Count, Sum, and Average functions in Pivot Tables.
4. I learned how to calculate and analyze the Mean and Median of sales data using Excel functions.
5. I learned how to analyze total sales, quantity sold, unit price, and profit using Pivot Tables and visual charts.
6. I improved my data analysis, visualization, and Excel skills through practical work.